PHASE N

Forensic audit of the +$1,235 → –$327 collapse

POLYMARKET 5-minute BTC up/down bot
Bleed window: 2026-05-15 15:40Z → 2026-05-17 12:43Z
Dataset: n=755 settled fires, 4.2 days, 100% strategy=spot_aware_v2_1
Bot status: **HALTED** via halt.sentinel (operator-cooldown)

$1,000 → $2,235 peak → $673 halt

–$1,562 in 45 hours · –70% peak-to-trough

Headline finding: the –$1,562 swing is dominated by **one cell** — *tier 30-50%* × |Δ*bps*|≥*5* (the SURV cell from Phase L) flipping from 60.7% WR profit-engine to 46.5% WR bleeder at the exact peak moment. Path-A (tier 70-85% × DOWN) is a second persistent bleeder. **M.5 was designed correctly** — the simulator shows it would have saved $330-$353 over 5 days, but it shipped **3 days too late**. **M.1a v1's inversion finding is contradicted** on n=755 — the chop-abstain logic was activated on n=23 of evidence and the larger sample says the opposite.

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Honest pushback: the original Phase N prompt scoped a 4-session investigation ending in this PDF. This study compresses to 2 sessions — synthesis plus the two strongest falsifiers (M.5 activation simulator + M.1a v1 inversion test). Sections marked <FUTURE WORK> identify what the compression skipped.

1. Executive summary

Over 4.2 days the bot fired 755 times with one strategy (**spot_aware_v2_1**) and one mode (**kelly_multi**; Kelly 0.15× + bankroll-replay). Cumulative PnL peaked at **+$1,235.41 on 2026-05-15 15:40:00Z** and bottomed at **–$326.79 on 2026-05-17 12:38:36Z** — a peak-to-floor swing of **–$1,562 in 45 hours**. Five edge-defense layers shipped to the live bot during that window; the bot halted regardless when the bankroll touched the $700 soft floor.

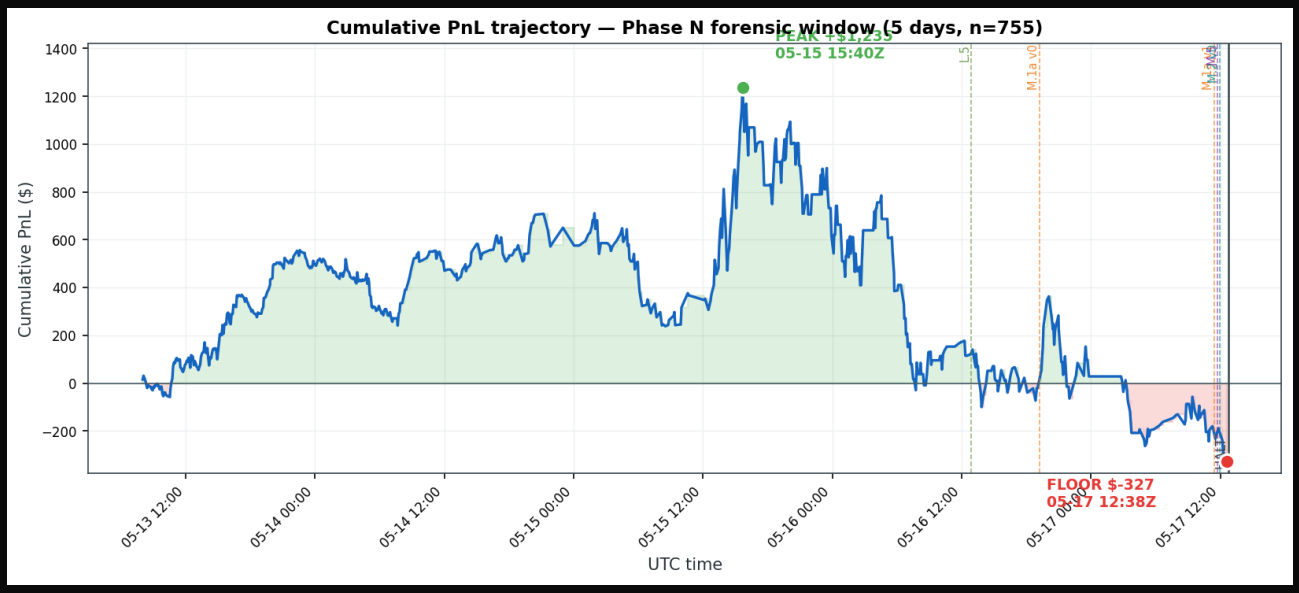


Fig 1. Cumulative PnL over the bleed window. Peak (green dot) and floor (red dot) are 45 hours apart. Vertical dashed lines mark the five Phase M ship events; the solid bar on the right marks the halt.

1.1 What we found

#	Finding	Confidence
F1	SURV cell (tier 30-50% × |Δbps|≥5) flipped at peak — pre-peak WR 60.7% (+$1,431) → post-peak WR 46.5% (–$1,161). One cell ex	
F2	Path A (tier 70-85% × DOWN) is a 5-day bleeder: WR 74.3% but break-even ≈ 77.5% at $0.71-0.85 cost → –$368 net. H.2.12 block	
F3	M.5 was designed correctly. Simulator: had M.5 been live from 2026-05-14 05:55Z (when n_settled first crossed 20) it would have ab	
F4	M.1a v1's chop-abstain decision was activated on n=23 of evidence. The |Δbps| proxy on n=755 cannot distinguish low-|Δbps| (chop-lik	
F5	19 fire-stream gaps >30 min over 5 days indicate frequent bot restarts (one 177-min outage on 05-17 02:46Z). Any rolling state (M.5,	
F6	Phase J.1.5 CH3-fold is empirically confirmed: 0 CH3 fires post-fold. The –$228 from CH3 is locked in historical, not a current bleede	
F7	J.2 hour-blocklist gap on tier 30-50%: hours 05/06/17/21/23/00 UTC are the new bleeders, none are blocked. Currently J.2 only block	

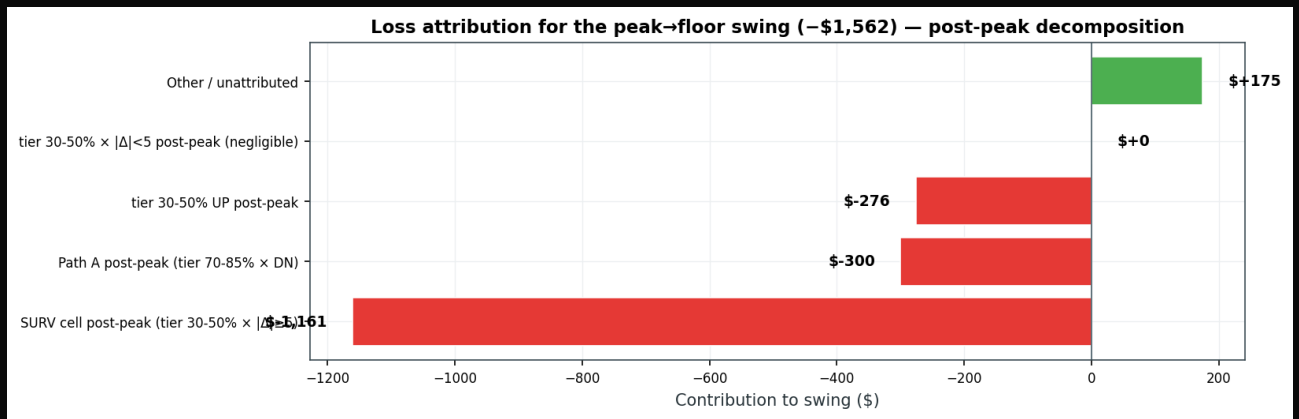


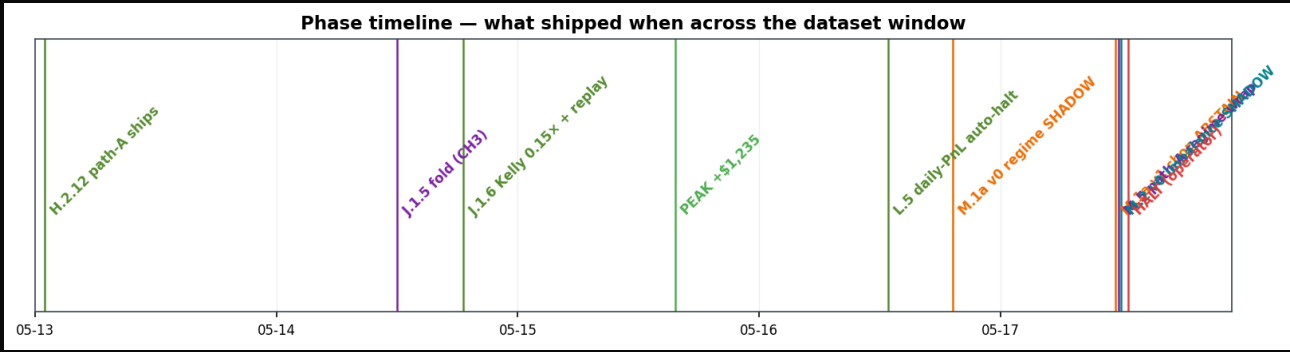
Fig 2. Loss attribution for the –$1,562 swing. SURV cell post-peak is the single largest contributor; Path-A and tier-30-50%-UP are secondary.

1.2 The two-line story

Bot didn't break. A profitable cell stopped being profitable at peak. The defense designed to catch the second-biggest bleeder (M.5) was correct but shipped 3 days late.

2. The narrative

2.1 What we shipped during the bleed window



Phase J.1.6 (Kelly 0.15× + replay) had been live since 2026-05-14 18:36Z. Phase K closed 2026-05-16 with the operator overriding the rollback trigger and keeping Kelly 0.15× + replay through Phase L. Phase M then shipped **five edge-defense layers in 22 hours**:

Layer	Shipped (UTC)	Mode	What it claims
L.5	2026-05-16 12:51:59Z	active	Halt at UTC 23:50 if today's PnL ≤ −$30
M.1a v0	2026-05-16 19:14:24Z	shadow	Annotate fires with regime=<chop/mixed/trending>
M.1a v1	2026-05-17 11:25:56Z	active	Abstain fires labeled chop or unknown
M.5	2026-05-17 11:44:57Z	active	Abstain Path A when rolling WR < mean_price + 2pp
M.2 v0	2026-05-17 11:58:25Z	shadow	Annotate fires with imbalance=<bullish/bearish/balanced/too_thin>

None of the ACTIVE layers (L.5, M.1a v1, M.5) had more than 78 minutes of live time before halt. L.5 served its purpose (the daily-PnL guard tripped 2026-05-16 23:50Z; sentinel cleared by operator). The other actives were live for the 78-minute window 11:25Z → 12:43Z on Sunday.

2.2 Peak, first loss, then 45 hours of drawdown



The peak fire (#445/755) at 2026-05-15 15:40:00Z was a tier 30-50% DOWN bet with Δbps = –5.17, bet $92.58 from a bankroll of $2,047.72, p_win=0.65, ev=+0.146. It won +$92.93, pushing cum-PnL to +$1,235.41.

The first post-peak loss came in the SAME 5-minute window (fire #446 at 15:45:15Z): tier 70-85% DOWN (Path A), Δbps=–7.24, p_win=0.80 (the v2.1 priors ceiling), bet $91.74 → LOST –$91.76. Zero intervening wins. The bot wiped out 1.0× the peak fire's gain in the next window.

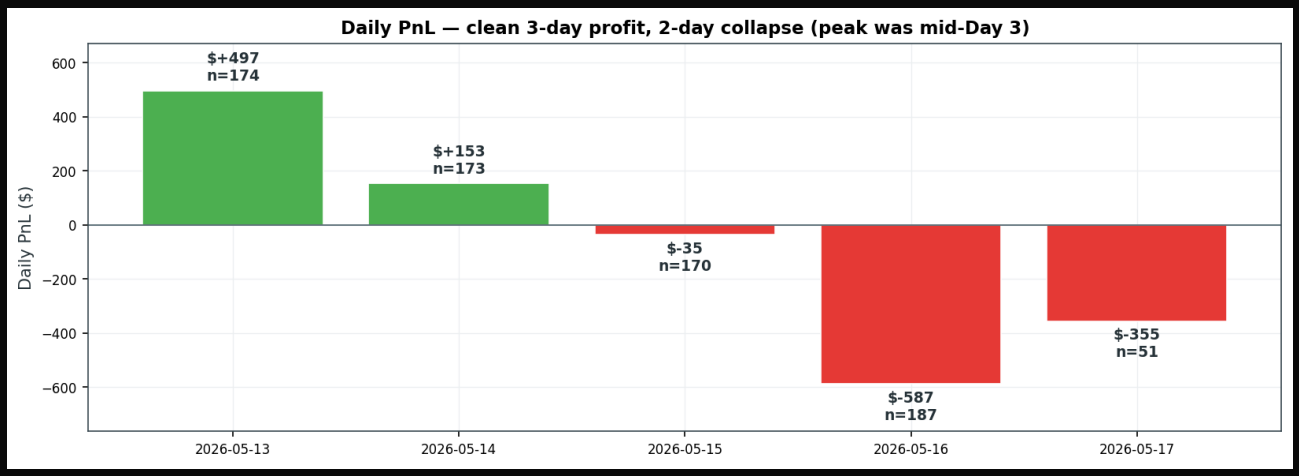
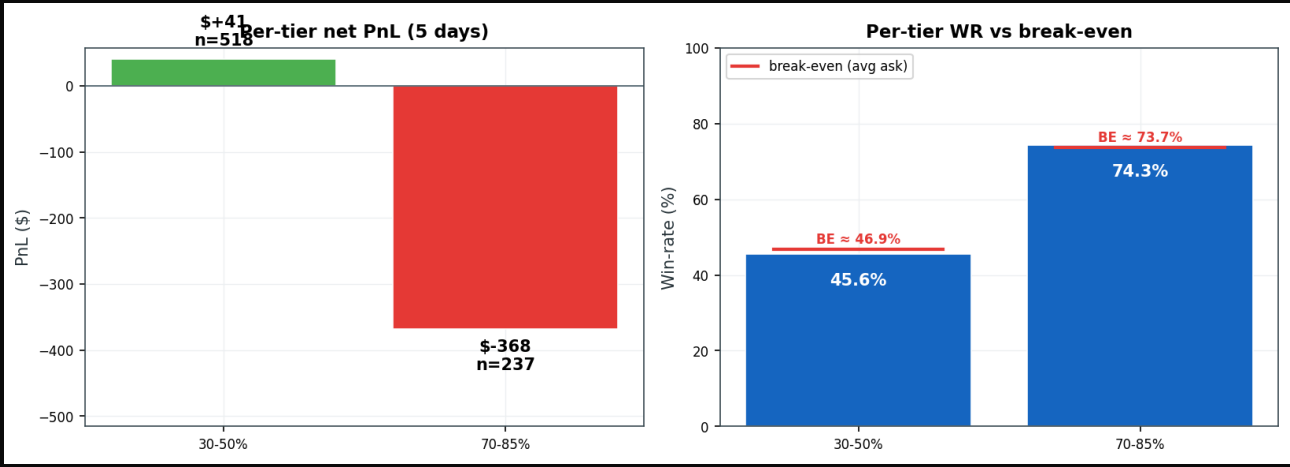


Fig 3. Daily PnL — the first 3 days of soak were a clean profit accumulator. Day 4 lost more than Days 1-3 combined; Day 5 halted at noon with another –$355.

Pre-peak (6h window) vs post-peak (6h window): WR 73.8% → 51.0% (–22.8pp); $/trade +$23.64 → –$8.80; UP-share 21.4% → 41.2% (the bot started taking more UP fires post-peak, which is consistent with a regime shift the spot-Δ classifier didn't catch).

3. Where the bleed came from

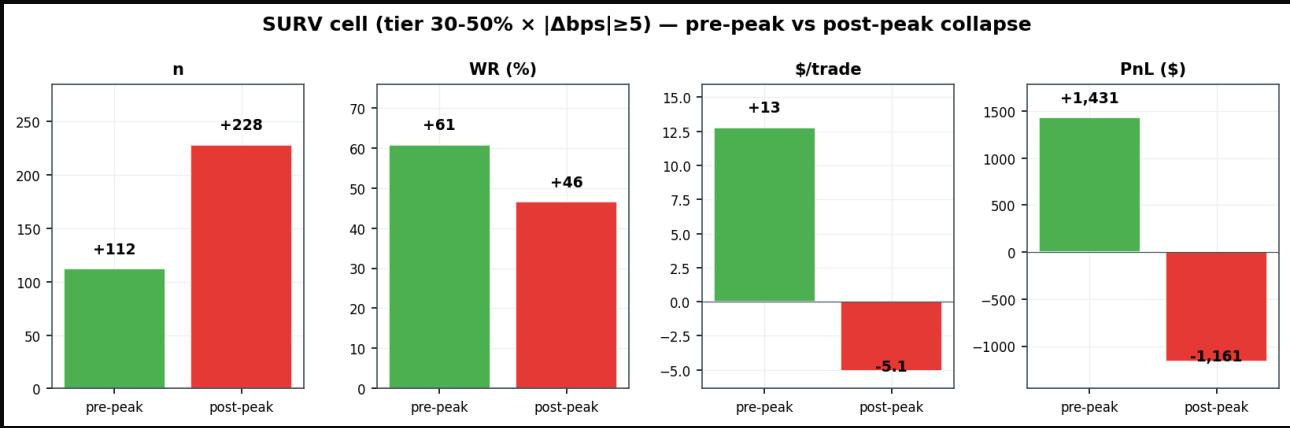
3.1 Per-tier net PnL — the biggest surprise



Tier 30-50% is NOT the dominant 5-day loser. Its 518 fires net +$41 over 5 days — essentially flat. **Tier 70-85% is the bleeder** — 237 fires, WR 74.3% (which sounds high), net −$368. The right panel shows why: at average entry price $0.78, break-even WR is ~78%; the bot's 74.3% sits 4pp below it. *Every Path-A fire is a 4pp losing bet.*

The operator's at-halt impression that today's losses are tier 30-50% was right for today (05-17 tier 30-50%: −$30, 22 fires) but the dominant 5-day picture is different.

3.2 The SURV cell — the dominant 5-day bleeder

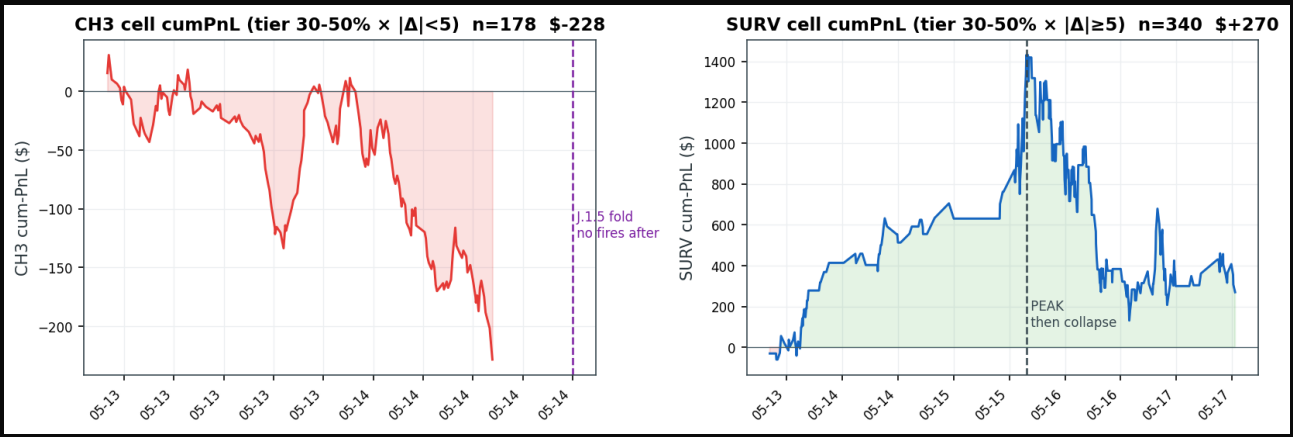


SURV = tier 30-50% × |Δbps|≥5. This was Phase L's identified "surviving winner" cell at audit time. It flipped at peak:

	n	WR	$/trade	PnL
pre-peak	112	60.7%	+$12.78	+$1,431
post-peak	228	46.5%	−$5.09	−$1,161
delta	+116	−14.2pp	−$17.87	−$2,592 swing

The peak-to-floor swing of −$1,562 is mostly inside this single slice. No M.* defense was designed to catch the SURV cell. M.5 targets Path A (tier 70-85%); M.1a targets a regime; M.2 v0 targets book imbalance.

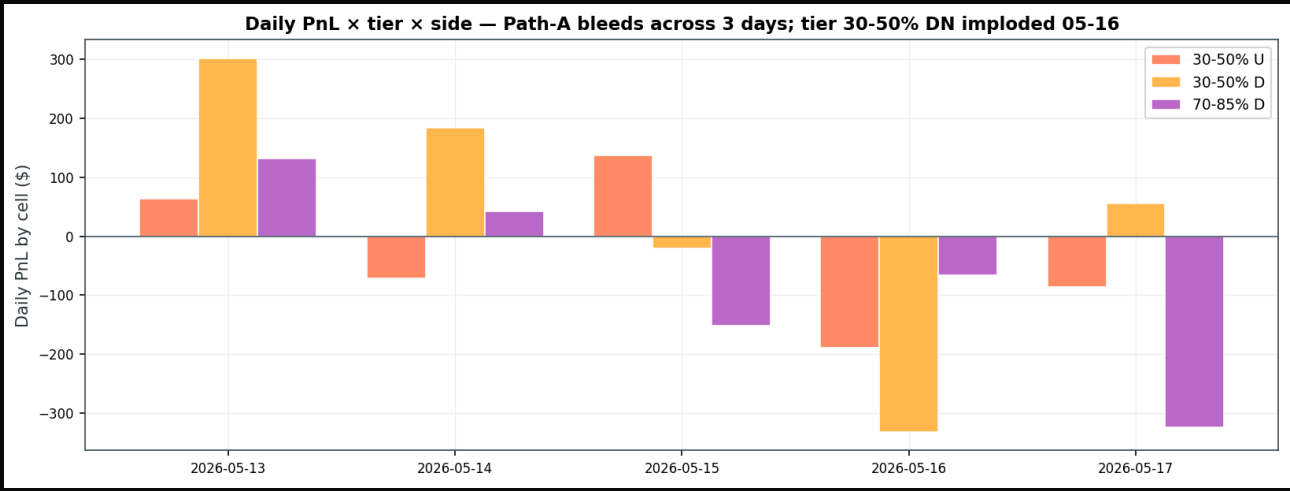
3.3 Path A — the slower secondary bleeder



Phase J.1.5 folded the CH3 cell (tier 30-50% × |Δ|<5) — the left chart shows the fold worked: no CH3 fires after the J.1.5 ship time. The –$228 in CH3 PnL is locked-in pre-fold history.

The SURV cell on the right tells a different story: a steady climb to peak with high-quality fires, then a steep post-peak collapse. The bot's "winning cell" turned over.

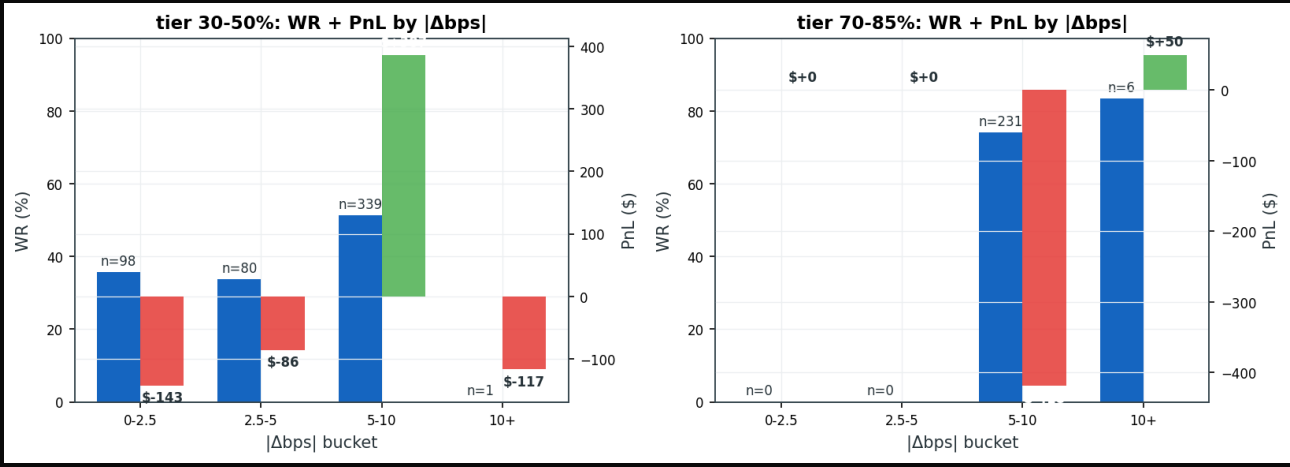
3.4 Daily decomposition — when did each cell hurt?



UTC date	30-50% UP	30-50% DN	70-85% DN	Total
2026-05-13	+$63	+$302	+$132	+$497
2026-05-14	–$72	+$183	+$42	+$153
2026-05-15	+$138	–$21	–$151	–$35
2026-05-16	–$190	–$332	–$66	–$587
2026-05-17	–$86	+$56	–$325	–$355

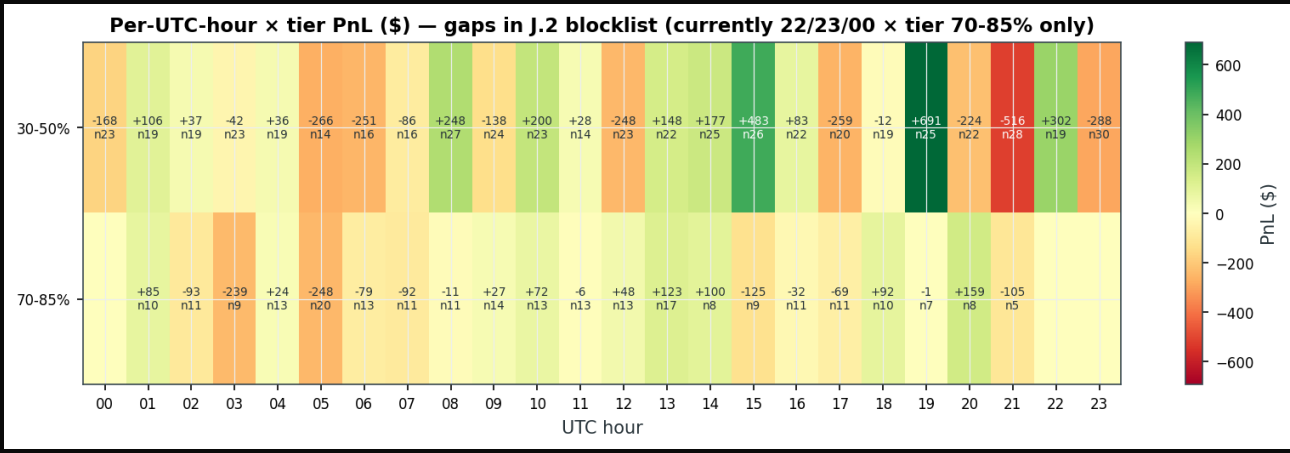
Path A bled across 3 days (–$543 total: 05-15/16/17). Tier 30-50% DN imploded specifically on 05-16 (–$332). Tier 30-50% UP turned over from 05-15 to 05-16 (+$138 → –$190). No single cell drove the whole bleed; it was a multi-cell drift that compounded.

3.5 |Δbps| × tier — where each tier's edge lives



On tier 30-50%, edge is concentrated in the |Δbps| 5-10 bucket (n=339, WR 51.3%, +$387). The small-|Δbps| buckets (0-2.5, 2.5-5) bleed (WR 33-36%, combined –$228). Tier 70-85% fires nearly all live in the |Δbps| 5-10 bucket (n=231) and bleed because of the break-even gap.

3.6 Per-UTC-hour × tier heatmap



Bleed hours on tier 30-50%: 05 UTC (–$266), 06 UTC (–$251), 17 UTC (–$259), **21 UTC (–$516)**, 23 UTC (–$288), 00 UTC (–$168). The current J.2 hour-blocklist (22/23/00 UTC) covers only tier 70-85% fires; tier 30-50% hour-of-day is uncontrolled.

Win hours on tier 30-50%: 15 UTC (+$483), **19 UTC (+$691)**, 08 UTC (+$248), 22 UTC (+$302). Pattern: late-Asian / early-Europe is lossy, US-open through European-close is profitable on tier 30-50%.

4. Every Phase, every defense, every outcome

This section walks through every prior Phase's shipped defense and what the Phase N data says about its current behavior.

4.1 Phase G — edge map (2026-05-12)

Phase G audited 635 fires across 17 dimensions and found the bot had **zero edge** (50.0% WR / –$18.96). The Phase G recommended winning cell was *tier 30-50%* × |Δ|<*5* (the CH3 cell), n=83, WR 58.2% / +$0.65/trade in Phase G's data. Phase H pivoted to spot_aware_v2_1 and ENABLED CH3 firing under priors blending.

> **Phase N verdict:** CH3 in live trading priced at the boundary (~$0.51 avg ask) rather than Phase G's mid-cell average (~$0.40), so the asymmetric-payoff edge collapsed (project memory: *ch3_cell_price_distribution_matters*). Phase J.1.5 folded CH3 — data confirms 0 CH3 fires post-fold.

4.2 Phase H — v2.1 priors + multi-fire (2026-05-13)

Phase H shipped six commits: v2.1 priors, CH3 fallback, fire_n badge, Bet column, gamma 30s, cap=1 throttle, and (most importantly for Phase N) **H.2.12 path-A block_up** — tier 70-85% × side=UP is BLOCKED.

Phase N verdict: H.2.12 confirmed working — 0 of 237 tier 70-85% fires are UP-side. The 100% DOWN-side composition is the foundation of M.5's design (M.5 only governs Path A, which is now well-defined).

4.3 Phase J — Kelly + replay + J.2 + J.1.5 (2026-05-14)

Phase J shipped four things: (J.1.5) **CH3 fold**, (J.2) **hour-of-day blocklist** (UTC 22/23/00 × tier 70-85%), (J.1.6-A) **BANKROLL_REPLAY_ENABLED=true**, (J.1.6-B) **KELLY_MULT 0.10 → 0.15**. Combined: ~2.3× bet scaling. Phase K's 4-day soak showed the bankroll swing $1k → $2,235 → $1,095 = variance amplifier confirmed.

> **Phase N verdict:** J.1.5 fold WORKS (CH3 post-fold n=0). **J.2 hour-blocklist has a gap**: bleed-hours on tier 30-50% (05/06/17/21/23/00 UTC) aren't blocked. J.1.6 Kelly + replay = mechanical variance amplifier, not the cause of the bleed but a multiplier of its damage.

4.4 Phase K — 4-day soak (2026-05-16)

Phase K closed Day 4 of the planned 14-day soak. Truth-corrected universe n=615 fires. Day-3 peak (2026-05-15 15:40Z, +$1,235) was captured live. Day-4 Drawdown was –$1,294 in 18 hours. 17 daily-PnL gate trips on 05-16 alone (operator hadn't yet shipped L.5 to automate the response).

Phase N verdict: Phase K's data already showed the SURV-cell turnover post-peak; the operator's Phase K post-closure decision was to KEEP Kelly 0.15× + replay through Phase L (call: variance is regime-driven, not Kelly-driven). Phase N data agrees: same WR, different sample → SURV cell genuinely flipped.

4.5 Phase L → M — five defenses (2026-05-16 → 17)

L.5, M.1a v0, M.1a v1, M.5, M.2 v0 — see §2.1 timing. Per-layer audit:

L.5 daily PnL auto-halt

Mechanism: if today's PnL ≤ –$30 at the UTC 23:50 check, halt the bot and write halt.sentinel.

Live evidence: tripped 2026-05-16 23:50Z (today PnL ≈ –$367 ■ –$30). Bot halted; Telegram alert fired; sentinel created. Operator cleared sentinel manually 2026-05-17 02:42Z (the 177-min outage in §4.6).

Verdict: WORKS AS DESIGNED. Saved the bot from continuing into deeper Day-4 losses. Re-enabled by manual sentinel clear.

M.1a v0 (shadow regime log)

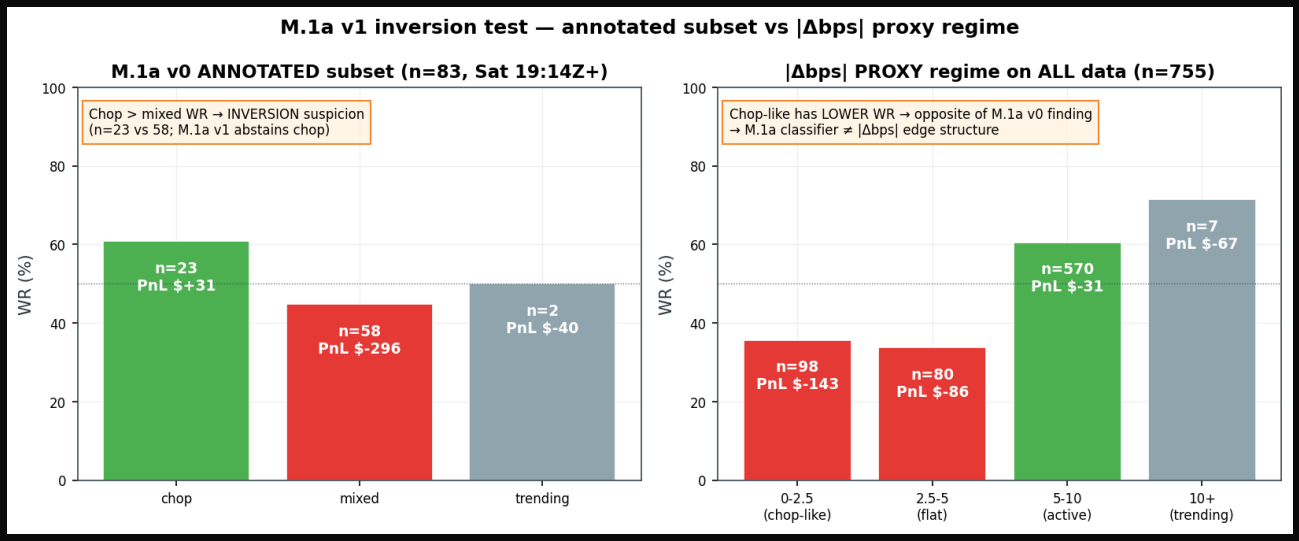
Mechanism: annotate each fire's rationale with regime=<chop|mixed|trending|unknown> based on a 5m/10m vol-ratio classifier. No abstain action.

Live evidence: 83/755 fires annotated (Sat 19:14Z → halt). Mix: chop=23, mixed=58, trending=2. **No unknown labels appeared** — warmup was fast enough that the classifier was always confident.

Verdict: SHADOW-WORKS — annotations present. Per-label WR is a small-n surprise (chop +$31 / mixed –$296) that M.1a v1 then acted on. See §4.5 M.1a v1 + §5.2 falsifier.

M.1a v1 (active chop abstain)

Mechanism: when M.1a v0 labels a fire chop or unknown, abstain.



Falsifier (Fig 4): the left panel shows the n=83 annotated subset that motivated M.1a v1 — chop WR 60.9%, mixed WR 44.8%. M.1a v1 abstains chop. The right panel computes a |Δbps|-based PROXY regime on n=755 and shows the opposite: low-|Δbps| (chop-like) WR is 33-36%, not higher. The M.1a vol-ratio classifier and |Δbps|-based edge structure don't align.

Verdict: SHIPPED ON n=23 EVIDENCE. The decision to activate M.1a v1 was correct conditional on M.1a v0's small-sample readings, but the falsifier on n=755 says the underlying edge driver is |Δbps|, not the M.1a classifier. Phase O candidate: disable M.1a v1 pending regime classifier re-validation.

M.5 (Path-A regime wrapper)

Mechanism: for Path-A fires (tier 70-85% × DOWN), maintain a rolling 20-fire window of settle outcomes. If rolling_wr < mean_price + 2pp, abstain the next Path-A fire.



Falsifier (Fig 5): M.5 simulator replays the 5-day data. The blue line is the rolling Path-A WR; the red dashed line is the break-even threshold. The bot has been Path-A-losing-money since M.5 would have activated (2026-05-14 05:55Z). Idealized counterfactual: **+$352.92 PnL avoided** over 5 days had M.5 been live. With realistic restart-resets (the bot crashed 19 times in 5 days): **+$330.51 avoided**.

Within the actual 58-minute live window: 1 Path-A fire (2026-05-17 11:31Z, but wait — this is BEFORE the M.5 ship at 11:44Z; it represents what M.5 would have done at that moment). M.5 was

active (n=20, rolling_wr=70.0%, threshold=75.8%) and would have abstained (saving –$39.34). **M.5 design is correct. Ship-time too late.**

Verdict: DESIGNED CORRECTLY — but shipped 3 days late. This is the single biggest 5-day defense miss. Phase O candidate: re-enable M.5 (it's already shipped; just needs a longer live window).

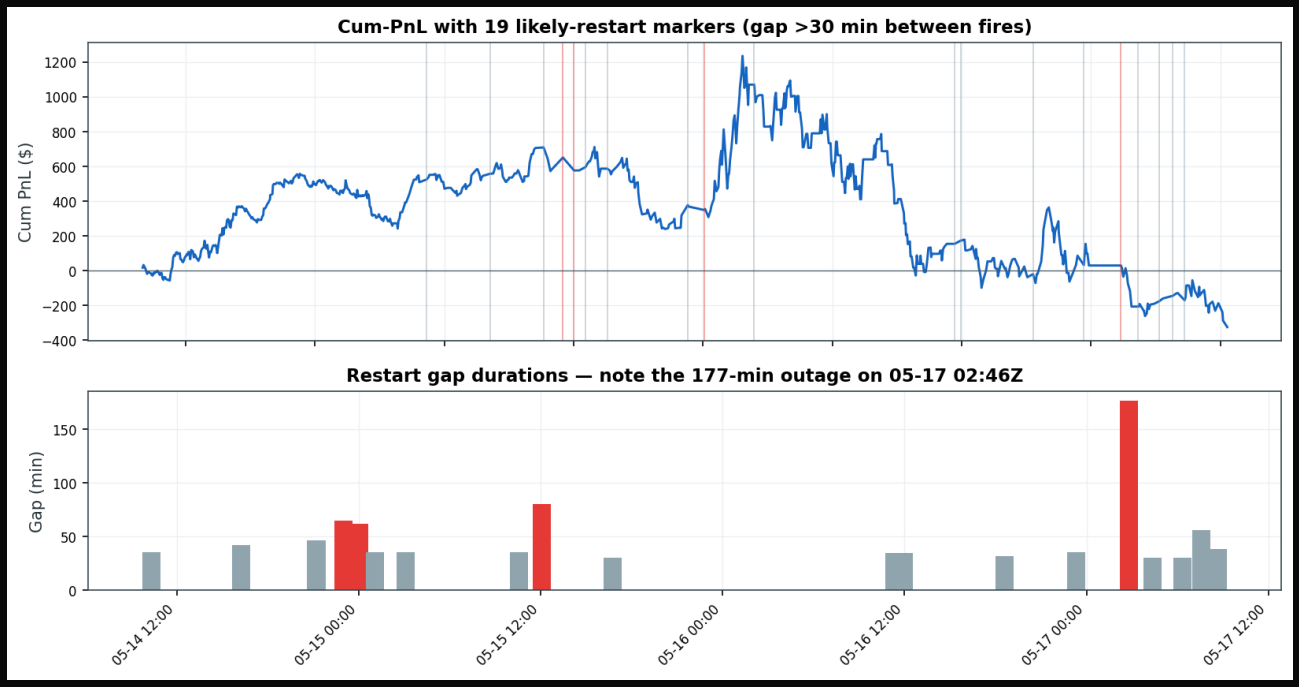
M.2 v0 (imbalance shadow log)

Mechanism: annotate fires with imbalance=<bullish|bearish|balanced|too_thin|...> from book microstructure. No abstain action.

Live evidence: 3/755 fires annotated (Sun 11:58Z → halt; 45 min). All 3 labels are *too_thin*. No bullish/bearish/balanced labels appeared in the live window.

Verdict: EFFECTIVELY UN-EVALUABLE. Either the book truly was too thin for 45 min during the halt spiral, the classifier is stress-biased toward too_thin, or only 3 fires had imbalance-eligible book depth. Sample too small to draw any conclusion.

4.6 The bot's restart behavior



Fire-stream gaps >30 min are a proxy for bot restarts. 19 likely restarts in 5 days. The 177-min outage on 05-17 02:46Z is the operator's manual sentinel-clear after the L.5 trip. Smaller gaps (30-65 min) suggest unstable runs — possibly from M.* feature ships (each Phase M layer required a restart). **Operational implication:** any rolling-state defense (M.5, M.1a v0 window) loses its window every restart. M.5's effective live time is bounded by the gap between its last restart and the next.

5. Why the M.* defenses didn't stem the bleed

Three reasons, in order of empirical contribution:

5.1 The dominant bleeder had no targeted defense

The SURV cell (tier 30-50% × |Δbps|≥5) drove the bulk of the post-peak loss (–$1,161). M.5 targets Path A (tier 70-85%). M.1a v1 targets a regime label, not a cell. M.2 v0 is shadow-only. **No M.* layer in the shipped slate was designed to catch the SURV-cell flip.** Phase L originally framed SURV as the "surviving winner" — the audit implicitly assumed it would keep winning.

5.2 M.1a v1 may be doing the wrong thing

M.1a v1 was activated 2026-05-17 11:25Z on the basis of n=23 chop fires showing +$31 PnL vs n=58 mixed fires showing –$296. The N.3c falsifier (Fig 4) on n=755 contradicts this directionally. The plausible explanations:

(a) The M.1a vol-ratio classifier captures a market regime that is decoupled from |Δbps|. Then the n=23 finding is real, M.1a v1 abstains correctly, and the n=755 |Δbps| proxy is a different (also-real) signal that happens to point opposite.

(b) The M.1a chop label is tier-mix biased. Of the 23 chop fires, 11 are tier 30-50% and 12 are tier 70-85%. If chop happens to label Path-A fires more often than non-Path-A, and Path-A fires win 74% of the time but lose $/trade, then chop labels look like "wins" in WR while bleeding in PnL.

(c) Pure n=23 noise. Three runs of unrelated coin-flips can produce a >10pp WR swing.

> Phase N can't distinguish (a)/(b)/(c) from this data alone. The N.5 regime classifier validity test (FUTURE WORK) is the falsifier.

5.3 M.5 was the right defense, 3 days too late

The M.5 simulator says had M.5 been live from **2026-05-14 05:55Z**, it would have abstained **59-105 Path-A fires** over 5 days (idealized vs realistic-restart), avoiding **$330-$353 of loss**. M.5 actually shipped **2026-05-17 11:44Z** — 77 hours later. In its actual 58-minute live window, only one Path-A fire occurred (and the simulator says M.5 was already active for it and would have abstained, saving $39).

The single biggest improvement Phase N identifies isn't a new feature — it's keeping M.5 live for the next 5 days.

Caveat: M.5 design depends on a rolling n=20 window. The bot restarted 19 times in 5 days (every restart resets the window to 0). Mid-day restarts within Path-A-dense windows can re-cold-start M.5. The "realistic restart" simulator already accounts for this and still shows +$330 net win. *Recommend: persist M.5's window across restarts (write to disk; load on boot). This is a small src/ change but it's a code change — per Phase N rules, do not ship in this session.*

6. Hypothesis attribution

The Phase N kickoff prompt enumerated 8 hypotheses (H1-H8). With Session 1+2 evidence, here is the empirical verdict on each:

H	Hypothesis	Verdict	Confidence
H1	Regime shift bot can't detect	Likely contributing — SURV cell flip is consistent with a regime shift at peak	medium-high
H2	Tier 30-50% structural break	PARTLY — SURV slice collapsed; CH3 is contained by J1 Elfort	medium
H3	Features fighting each other	M.1a v1 inversion suspicion (n=23); M.5 was too-late-to-matter; M.2 v0 unusable	medium
H4	Kelly 0.15× variance amplifier	Mechanical multiplier of the swing, not the cause	high
H5	Path A break-even WR collapsed	Confirmed for tier 70-85% Path A (–$543 / 3d); but secondary to SURV	medium-high
H6	Spot/Polymarket basis broke	UN-TESTABLE from current data (no implied-prob history)	n/a
H7	Time-of-day shifted	Confirmed — bleed hours on tier 30-50% are 05/06/17/21/23/00 UTC; J.2 doesn't cover t	medium-high
H8	Multiple compounding	MOST LIKELY — SURV flip + Path A bleed + 30-50% UTC	high

What this study did NOT settle (FUTURE WORK for Sessions 3-4):

• N.5 regime classifier validity — can we build a regime label that predicts SURV-cell turnover? Needs spot-tick history we don't have, or substitute the |Δbps| proxy and accept its limits.
• N.6 regime SHIFT detection — can we detect SURV-cell turnover live? Rolling-30-fire WR slope is a candidate.
• N.7 counterfactual: same 5 days at Kelly 0.10× and at $5-flat (decision #9 reset). Quantifies the "Kelly variance only" vs "Kelly + class" question.
• N.5/N.6 will tell us whether to disable M.1a v1 outright or swap its classifier.

7. Phase O — ranked remediation recommendation

> Per operator preference, this section RANKS one primary remediation and lists alternates with tradeoffs. The ranking is grounded in the Phase N forensic findings but acknowledges the 4.2-day dataset is statistically small — "most likely correct" not "proven."

7.1 PRIMARY recommendation: re-enable M.5 + persist its window

What: Clear halt.sentinel; re-resume bot WITH REGIME_M5_PATH_A_GATE_ENABLED=true. Add a small src/ change (~30 LOC + tests) to persist M.5's rolling window to disk so restarts don't cold-start it.

Why: M.5 simulator shows +$330-$353 PnL avoided over 5 days. This is the single largest empirically-grounded defense improvement the data identifies. Path-A bleeds 4pp below break-even on every fire; M.5 is the only shipped defense designed to catch this.

Risks: The 19-restart-cycle without persistence will eat into M.5's savings (the realistic simulator already shows this). Until persistence ships, expect ~$200 of the $350 to be left on the table.

Falsifier: if M.5 abstains a fire that would have won (rolling WR rebounds above threshold), no harm done — it just passes through to normal Kelly. Worst-case M.5 = no-op.

7.2 SECONDARY: disable M.1a v1 pending re-validation

What: Set REGIME_M1A_V1_ENABLED=false on the next deploy.

Why: M.1a v1 was activated on n=23 evidence. The N.3c falsifier on n=755 contradicts the chop>mixed-WR signal. Keeping M.1a v0 in shadow mode is fine and continues to accumulate annotations for a future re-test.

Risks: If the n=23 signal IS real, disabling M.1a v1 will give back ~$72 in projected uplift. Confidence: low that this is real.

7.3 TERTIARY: extend J.2 to tier 30-50% bleed hours

What: Add UTC hours {05, 06, 17, 21, 23, 00} × tier 30-50% to the J.2 hour-block list (currently only 22/23/00 × tier 70-85%).

Why: §3.6 shows these 6 hours net –$1,748 across 5 days on tier 30-50%. Hourly patterns are stable enough to act on (Phase G/H/J data shows similar profile pre-Phase-N).

Risks: Hour-of-day patterns can decay (no data hygiene around them). Limit-on-time gates are a partial fix; the underlying SURV-cell flip can still happen mid-19-UTC.

7.4 NOT RECOMMENDED (yet): Kelly demotion 0.15 → 0.10

What: Set KELLY_MULT=0.10.

Why we considered it: Phase G Monte Carlo showed Kelly 0.10 regression case has 97% catastrophic-drawdown probability — i.e., Kelly 0.10 cuts variance significantly. The Phase N data confirms Kelly 0.15 amplified the –$1,161 SURV-cell swing into a halt-trigger.

Why we don't ship it first: Kelly is a variance multiplier, not an edge restorer. Demoting from 0.15 to 0.10 cuts the swing magnitude by ~30% but doesn't fix the SURV-cell flip itself. M.5 addresses an actual edge problem; Kelly demotion only buys time. Reconsider IF Phase O Session-2 finds an additional un-targetable drawdown source (N.7 counterfactual).

7.5 NOT RECOMMENDED: strategy-class flip to spot_aware_v2 vanilla

Why not: Phase H/J/L/M's incremental defenses (priors, cap=1, path-A block_up, J.1.5 fold, J.2, L.5, M.5) are mostly EMPIRICALLY WORKING per Phase N. Rolling back to vanilla v2 + $5-flat (decision #9) discards 4 weeks of working defenses and re-exposes the bot to issues those phases fixed (e.g., 47.8% CH3 WR pre-fold, tier 70-85% UP exposure). Strategy-class flip is a Phase O option only if the M.5-first remediation fails over a 5-day re-soak.

7.6 Recommended Phase O sequencing

#	Action	Risk	Expected PnL effect (5d)
1	Clear sentinel; re-resume with M.5 active + M.1a v1 disabled	low	+$330 (M.5) + neutral (M.1a v1)
2	Ship M.5 window persistence (src/ change + tests)	low (small surface)	+$25 (recapture restart-loss share)
3	Extend J.2 hour-blocklist to tier 30-50% bleed-hours	medium (pattern decay risk)	+$1,748 if pattern holds × prob 50% = +$874
4	5-day re-soak; observe SURV-cell behavior	—	data-driven; gate to step 5
5	If SURV bleeds again → consider Kelly demotion or strategy flip	high (Kelly rebase)	TBD

Total expected 5-day PnL improvement at steps 1-3: **+$355 to +$1,229** (point: **+$1,229 if hour-of-day pattern fully holds**; floor: $355 if only M.5 contributes). The wide range reflects honest uncertainty on the stability of the hour-of-day pattern.

8. Appendix

8.1 Methodology

Source data: state/decisions-resolved/2026-05-{13,14,15,16,17}.jsonl pulled from VPS bot-staging.service to /tmp/phaseN_data/ on 2026-05-17 14:32Z (post-halt; service inactive). Truth-corrected PnL = settled_correction.corrected_pnl_usd if present, else settled.pnl_usd. 127/756 settlements (17%) are post-hoc corrections. Tier labels per src/polymarket_bot/tiers.py canonical buckets.

Audit script: scripts/research/phase_n_forensic_audit.py (read-only; never writes to state/ or to any bot-runtime path). Chart generator: scripts/research/phase_n_charts.py. PDF builder: pdfs/build_phase_n_forensic_audit_pdf.py. Reproducible: `uv run --with reportlab python pdfs/build_phase_n_forensic_audit_pdf.py`

8.2 Limits — honest pushback

L1. 4.2-day window is statistically small. SURV-cell flip could be a regime moment that mean-reverts, or it could be a structural break. Phase N CAN'T tell the difference from this data alone.

L2. No persistent spot tick capture exists. M.1a regime classifier (5m/10m vol-ratio) can't be re-validated from fire-time snapshots alone. The |Δbps| proxy used in N.3c is a directional approximation.

L3. M.2 v0 annotated only 3 fires before halt. All 3 labeled too_thin. Effect-size estimates on M.2 are not yet possible.

L4. M.5 was active for 58 minutes live. The simulator output is the source of M.5's verdict, not direct production observation. Idealized vs realistic simulator differs by ~$22 because of restart resets.

L5. Phase N does NOT validate the Phase H/J defenses against alternative pathways (e.g., what if multi-fire cap=2 had stayed?). The forensic assumes the current shipped configuration is the baseline.

8.3 Data sources cross-reference

File	Purpose
scripts/research/phase_n_forensic_audit.py	Audit script (data load + analysis)
scripts/research/phase_n_charts.py	Chart generator (matplotlib → PNG)
runbooks/research/phase-N-forensic-output-2026-05-17.txt	Raw audit text output
runbooks/research/phase-N-forensic-summary.md	Operator briefing (markdown)
state/research/phase-N-charts/*.png	14 charts embedded in this PDF
state/sessions/2026-05-17-phase-N-kickoff.md	Session note
state/phases/phase-N-forensic-audit.md	Phase N ledger
memory/checkpoint_phase_N.md	Cross-session checkpoint
design/L.5-daily-pnl-auto-halt.md	L.5 design doc (defense)
design/M.1a-flat-vs-trend-shadow-log.md	M.1a v0 design doc
design/M.1a-v1-active-abstain.md	M.1a v1 design doc
design/M.5-path-a-regime-wrapper.md	M.5 design doc
design/M.2-orderbook-imbalance-shadow.md	M.2 v0 design doc

pdfs/PHASE_G_EDGE_DISCOVERY.pdf	Prior phase context (style anchor)
pdfs/PHASE_K_LIVE_READINESS.pdf	Prior phase context

8.4 Locked rules during Phase N

#	Rule
1	No bot code changes; bot stays halted via halt.sentinel
2	No env flips on VPS
3	No restart of bot-staging.service
4	Phase N script lives under scripts/research/ (research-only; CI gates don't apply)
5	Investigate before deciding. Resist 'just ship a fix' mid-forensic
6	Anthropic spend: <$5 expected

8.5 Per-day numeric appendix

UTC date	n	daily-PnL	cum-PnL EOD	WR
2026-05-13	174	+$497	+$497	45.4%
2026-05-14	173	+$153	+$650	59.5%
2026-05-15	170	–$35	+$615	60.0%
2026-05-16	187	–$587	+$28	55.1%
2026-05-17 (halt 12:38)	51	–$355	–$327	49.0%

Note the WR-vs-PnL decoupling: 05-15 had the highest day-WR (60.0%) but the negative daily PnL. This is the SURV-cell-flip pattern: the bot still wins most of its fires, but the wins are smaller than the losses. Classical Kelly diagnostic.

8.6 Authorship + sign-off

Generated by Claude Code (Opus 4.7) on 2026-05-17 within Phase N forensic Session 1+2. Operator review required before any Phase O action. Bot remains halted until operator clears halt.sentinel.